

Yohannes Alelign Biresaw

Research Profile | Secure and Robust Federated Learning

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Focused research direction

My research direction is the security, privacy, and robustness of distributed machine-learning systems. I am particularly interested in secure federated learning, adversarial clients, robust aggregation, trustworthy evaluation under non-IID data, and privacy-aware deployment.

My completed federated medical-imaging project is the strongest evidence of my current research preparation. It provides a practical foundation for a future thesis on how a federated system can distinguish malicious updates from legitimate variation caused by heterogeneous client data.

Completed research project

Evaluating FedAvg and FedProx for Federated Chest-Radiograph Classification Under Non-IID Client Heterogeneity

Using 26,684 RSNA examinations linked to 11,452 patients, I created patient-exclusive data splits, simulated five federated clients, compared centralized learning, FedAvg, and FedProx under IID and non-IID settings, repeated experiments across three seeds, and used a frozen final-test protocol.

Condition	ROC-AUC	PR-AUC	Balanced accuracy	F1-score
Centralized	0.8295 $\pm$ 0.0040	0.5800 $\pm$ 0.0057	0.7510 $\pm$ 0.0037	0.5745 $\pm$ 0.0107
FedAvg IID	0.8179 $\pm$ 0.0030	0.5605 $\pm$ 0.0042	0.7453 $\pm$ 0.0048	0.5695 $\pm$ 0.0068
FedAvg moderate non-IID	0.8131 $\pm$ 0.0027	0.5553 $\pm$ 0.0065	0.7355 $\pm$ 0.0032	0.5541 $\pm$ 0.0055
FedAvg severe non-IID	0.8070 $\pm$ 0.0005	0.5440 $\pm$ 0.0043	0.7312 $\pm$ 0.0020	0.5529 $\pm$ 0.0029
FedProx moderate non-IID	0.8112 $\pm$ 0.0020	0.5505 $\pm$ 0.0040	0.7375 $\pm$ 0.0050	0.5608 $\pm$ 0.0036
FedProx severe non-IID	0.8076 $\pm$ 0.0019	0.5428 $\pm$ 0.0092	0.7335 $\pm$ 0.0030	0.5581 $\pm$ 0.0076

Stronger heterogeneity reduced federated performance. FedProx reduced client and global update magnitudes by roughly one-half, but it did not consistently improve held-out ranking. This distinction between optimization stability and predictive generalization motivated my security-focused next step.

Proposed MSc direction

Secure and Robust Federated Learning Under Heterogeneous and Adversarial Clients

The proposed research will study how robust aggregation can reject model-poisoning updates without excluding honest clients whose updates are unusual because of non-IID data. Evaluation will include malicious-client attacks, robust aggregation baselines, benign-client false rejection, worst-client performance, attack success, calibration, and computational cost.

Proposed research questions

- How does legitimate label or acquisition heterogeneity change the geometry of honest client updates?
- Which robust aggregators preserve minority-client utility while limiting model-poisoning attacks?
- How often do security defenses reject honest clients because their data are different rather than malicious?

Research preparation

- Python, TensorFlow/Keras, NumPy, pandas, scikit-learn, Git, and GitHub.
- DICOM preprocessing, CNN development, class-imbalance evaluation, and threshold selection.
- FedAvg, FedProx, IID/non-IID partitioning, and client-drift analysis.
- Patient-aware design, multi-seed evaluation, frozen-test controls, and reproducibility documentation.
- Professional experience in IT support, networking, Windows systems, and practical troubleshooting.

Selected academic projects

- Obstacle-avoidance robotic vehicle for industrial applications: simulation-based Arduino motor and sensor control; final-year project with a 30-40-page report.
- Agricultural quadcopter for crop and seed-health monitoring: simulation-based Arduino motor and camera control.

What I am seeking

A funded thesis-based MSc opportunity with a supervisor working on federated learning, adversarial machine learning, trustworthy AI, privacy-preserving machine learning, or distributed-system security. I am prepared to strengthen my mathematical and security foundations while contributing a reproducible experimental pipeline and careful empirical analysis.

Research strengths

- Ability to connect experimental design with security questions rather than treating accuracy as the only outcome.
- Careful handling of patient leakage, validation selection, class imbalance, and reproducibility controls.
- Practical systems background that supports implementation, troubleshooting, and research infrastructure.
- Willingness to report mixed or negative findings without overstating conclusions.

Potential contribution to a research group

I can contribute a working federated medical-imaging pipeline, verified experiment records, and a disciplined approach to evaluation. Under supervision, I aim to deepen the mathematical and security analysis, implement robust aggregation and attack models, and develop the work into a thesis and publishable study.

GitHub: <https://github.com/john121319/federated-pneumonia-research>